Ramakrishnan K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.93
SRI RAMAKRISHNA MATRIC HIGHER SECONDARY SCHOOL	Class XII (TAMILNADU STATE BOARD)	2014	91.83%
SRI RAMAKRISHNA MATRIC HIGHER SECONDARY SCHOOL	Class X (TAMILNADU STATE BOARD)	2012	93.20%

TECHNICAL SKILLS & RELEVANT COURSEWORK

Programming & Data: Python, C++, SQL, NumPy, Pandas, SciPy

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Regression, ANN, CNNs, Vision Transformers (ViT), NLP, LLMs, RAG, PINNs, Surrogate Modeling

Optimization & Analytics: NSGA-II/III, MILP, Differential Evolution, Sobol Sensitivity Analysis, Multi-Objective Optimization

Tools: OpenCV, Git, GitHub, MATLAB, ANSYS, COMSOL

AI/ML & Data Science: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Reinforcement Learning, Statistical Analysis, Physics-Informed Modeling, Optimization Techniques.

Energy & Engineering: Battery Technologies, Electric Vehicles, Photovoltaic Technology, Electrochemistry, Renewable Energy Systems, Energy Management.

Mathematics: Probability Theory, Bayesian Statistics, Linear Algebra, Convex Optimization, Differential Equations.

M.TECH THESIS

Development of Latent-Space Generative Surrogate Models for Accelerated Combustion CFD Prediction
(Ongoing)

- Developing a latent-space surrogate modeling framework to predict combustion temperature fields from operating conditions, including pressure, ammonia concentration, and air preheat temperature.
- Implementing CNNs and Vision Transformers (ViT) for latent representation learning and combustion field reconstruction, with Swin Transformers as an extension.
- Exploring Physics-Informed Neural Networks (PINNs) to incorporate physical constraints and enhance the physical consistency of predicted combustion fields.

TECHNICAL PROJECTS

1. LLM-Based Literature Review | Python, NLP, LLMs

- Developed an LLM-assisted literature review workflow to analyze AI/ML applications in ammonia combustion using Scopus-indexed research publications.
- Performed literature screening, data preprocessing, and thematic clustering using Python, Pandas, Sentence Transformers, UMAP, and HDBSCAN.
- Leveraged Google Gemini and NotebookLM for thematic analysis, literature synthesis, and research-gap identification.

2. Multi-Objective Engineering Optimization | Python, NSGA-II

- Formulated a multi-objective engineering optimization problem involving design variables, competing objectives, and constraints.
- Implemented NSGA-II to explore objective trade-offs and identify Pareto-optimal solutions.
- Analyzed Pareto fronts and optimization results to evaluate trade-offs and support engineering design decisions.

POSITIONS OF RESPONSIBILITY (POR)

- Placement Coordinator | Greenko School of Sustainability, IIT Hyderabad
- Teaching Assistant | IIT Hyderabad
- Sunshine PG Mentor | IIT Hyderabad